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🌟 Luminer: AI Teaching Assistant - General Q&A Mode

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Show that a single-mode thermal state of light is first-order coherent but not second- or higher-order coherent.

Hello! I am Luminer, your AI teaching assistant for General Physics. I'd be glad to help you dive into the quantum optics of thermal light!

To show that a single-mode thermal state is first-order coherent but not second- or higher-order coherent, we need to apply Roy Glauber's quantum theory of optical coherence.

Let's break down the core physical concepts and the mathematical derivation step by step.

1. The Single-Mode Thermal State

A thermal state of light in a single mode is a statistical mixture of photon number states (also known as Fock states). Let the mean photon number be $\bar{n}$. The density operator ρ for a thermal state is given by:

$$\rho = \sum_{n=0}^{\infty} P(n)|n\rangle\langle n|$$

where the probability $P(n)$ of finding n photons follows the Bose-Einstein distribution:

$$P(n) = \frac{\bar{n}^n}{(1+\bar{n})^{n+1}}$$

2. Definition of Quantum Coherence

In quantum optics, the normalized m-th order coherence function at equal times (which is standard when analyzing the photon statistics of a single-mode stationary field) is defined using the creation operator $a^\dagger$ and annihilation operator a:

$$g^{(m)} = \frac{\langle (a^\dagger)^m a^m \rangle}{\langle a^\dagger a \rangle^m}$$

A field is defined as m-**th order coherent** if and only if $|g^{(k)}| = 1$ for all integers $k \leq m$.

3. Proof of First-Order Coherence

First-order coherence typically includes a time delay τ between the two measurements. The normalized first-order coherence function is:

$$g^{(1)}(\tau) = \frac{\langle a^\dagger(t) a(t+\tau) \rangle}{\langle a^\dagger(t) a(t) \rangle}$$

For a single-mode field oscillating at an angular frequency ω, the time evolution of the annihilation operator is:

$$a(t) = ae^{-i\omega t}$$

Substituting this into the numerator, we get:

$$\langle a^\dagger(t) a(t+\tau) \rangle = \langle a^\dagger e^{i\omega t} a e^{-i\omega(t+\tau)} \rangle$$

$$\langle a^\dagger(t)a(t+\tau)\rangle = \langle a^\dagger a\rangle e^{-i\omega\tau}$$

The denominator is simply the average photon number:

$$\langle a^\dagger a\rangle = \bar{n}$$

Therefore, the first-order coherence function becomes:

$$g^{(1)}(\tau) = \frac{\bar{n}e^{-i\omega\tau}}{\bar{n}} = e^{-i\omega\tau}$$

Taking the absolute magnitude gives:

$$|g^{(1)}(\tau)| = 1$$

Conceptual Breakdown: Because $|g^{(1)}(\tau)| = 1$, the single-mode thermal state is **perfectly first-order coherent**. Physically, this means that if you split this light and recombine it (for example, in a Michelson interferometer), it will produce perfect interference fringes. This happens because a single mode contains exactly one definite frequency, so it behaves like a perfect continuous sine wave in terms of its phase relationship with itself over time.

4. Proof of Second- and Higher-Order Incoherence

Next, we calculate the normally ordered expectation value $\langle (a^\dagger)^m a^m\rangle$ for the thermal state. Applying the annihilation operator m times to a Fock state $|n\rangle$ yields:

$$a^m|n\rangle = \sqrt{\frac{n!}{(n-m)!}}|n-m\rangle$$

The expectation value for a single pure Fock state is therefore:

$$\langle n|(a^\dagger)^m a^m|n\rangle = \frac{n!}{(n-m)!}$$

For the statistical mixture of the thermal state, we take the ensemble average by summing over all possible photon numbers $n \geq m$:

$$\langle (a^\dagger)^m a^m \rangle = \sum_{n=m}^{\infty} P(n) \frac{n!}{(n-m)!}$$

Substitute the thermal probability distribution $P(n)$ into the equation:

$$\langle (a^\dagger)^m a^m \rangle = \sum_{n=m}^{\infty} \left[\frac{\bar{n}^n}{(1+\bar{n})^{n+1}} \right] \frac{n!}{(n-m)!}$$

To evaluate this sum, we introduce a change of index $k = n - m$, which means $n = k + m$:

$$\langle (a^\dagger)^m a^m \rangle = \sum_{k=0}^{\infty} \frac{\bar{n}^{k+m}}{(1+\bar{n})^{k+m+1}} \frac{(k+m)!}{k!}$$

We can factor out the terms that do not depend on the summation index k:

$$\langle (a^\dagger)^m a^m \rangle = \frac{\bar{n}^m}{(1+\bar{n})^{m+1}} \sum_{k=0}^{\infty} \left(\frac{\bar{n}}{1+\bar{n}} \right)^k \frac{(k+m)!}{k!}$$

Now, we utilize the standard Taylor series expansion for a negative binomial, which is:

$$\sum_{k=0}^{\infty} x^k \frac{(k+m)!}{k!} = m!(1-x)^{-(m+1)}$$

In our equation, we let the variable x be:

$$x = \frac{\bar{n}}{1+\bar{n}}$$

Consequently, the term $(1-x)$ becomes:

$$1 - x = 1 - \frac{\bar{n}}{1+\bar{n}} = \frac{1}{1+\bar{n}}$$

Thus, the factor $(1-x)^{-(m+1)}$ evaluates to:

$$(1-x)^{-(m+1)} = (1+\bar{n})^{m+1}$$

Substituting this mathematical identity back into our expectation value calculation:

$$\langle (a^\dagger)^m a^m \rangle = \frac{\bar{n}^m}{(1+\bar{n})^{m+1}} \cdot m! \cdot (1+\bar{n})^{m+1}$$

The terms $(1+\bar{n})^{m+1}$ cancel each other out, leaving us with a wonderfully simple result:

$$\langle (a^\dagger)^m a^m \rangle = m!\bar{n}^m$$

Finally, we plug this result back into the definition of the m-th order coherence function $g^{(m)}$:

$$g^{(m)} = \frac{m!\bar{n}^m}{(\bar{n})^m}$$

$$g^{(m)} = m!$$

Conceptual Breakdown: For $m = 2$, we have $g^{(2)} = 2$. For $m = 3$, $g^{(3)} = 6$, and so forth. Because $g^{(m)} \neq 1$ for any $m \geq 2$, the thermal state is **not** second- or higher-order coherent!

Physically, a value of $g^{(2)} = 2$ indicates a phenomenon called **photon bunching**. Unlike a coherent laser beam (which has a steady intensity and $g^{(2)} = 1$), a thermal light source has massive intensity fluctuations. Photons from thermal light tend to arrive at a detector in "bunches" together rather than being spaced out randomly. This property was famously observed in the historic Hanbury Brown and Twiss intensity correlation experiment.

I hope this clarifies the mathematics and the fascinating physics behind the coherence of thermal states! Please let me know if you would like to explore any of these steps in more detail.

+ Enter your physics question here: